Machine Learning-based Smart Helmet for Personalized Construction Monitoring

Progress Presentation - 01



Project ID: R26-SE-042

Problem & Impact

- Construction workers are exposed to **various hazardous conditions**, making construction one of the **most dangerous industries**.
- **Traditional safety systems** are largely reactive, heavily depend on human judgement, and fail to detect real-time risks.
- To overcome the limitations of traditional safety approaches, **wearable sensor devices** have emerged as a promising technology for continuous safety monitoring.
- Existing wearable solutions,
  - **Issue 01:** Monitor environmental and physiological conditions separately
  - **Issue 02:** mainly relies on fixed threshold values, resulting in incomplete and less accurate risk assessments.
- These limitations can lead to delayed hazard detection, false alarms, workplace accidents, productivity loss, and increased operational costs.





Novelty & User Persona

- Personalized AI-driven worker safety monitoring framework for construction environments.
  - **Combines environmental and physiological monitoring** into a single integrated risk assessment system.
  - Uses **worker-specific physiological baseline analysis** instead of fixed threshold values.
  - **Machine learning-based classification** of worker states as Safe, Warning, or Critical.
  - Real-time alerts, adaptive monitoring, and AI-powered safety recommendations.



Construction Workers

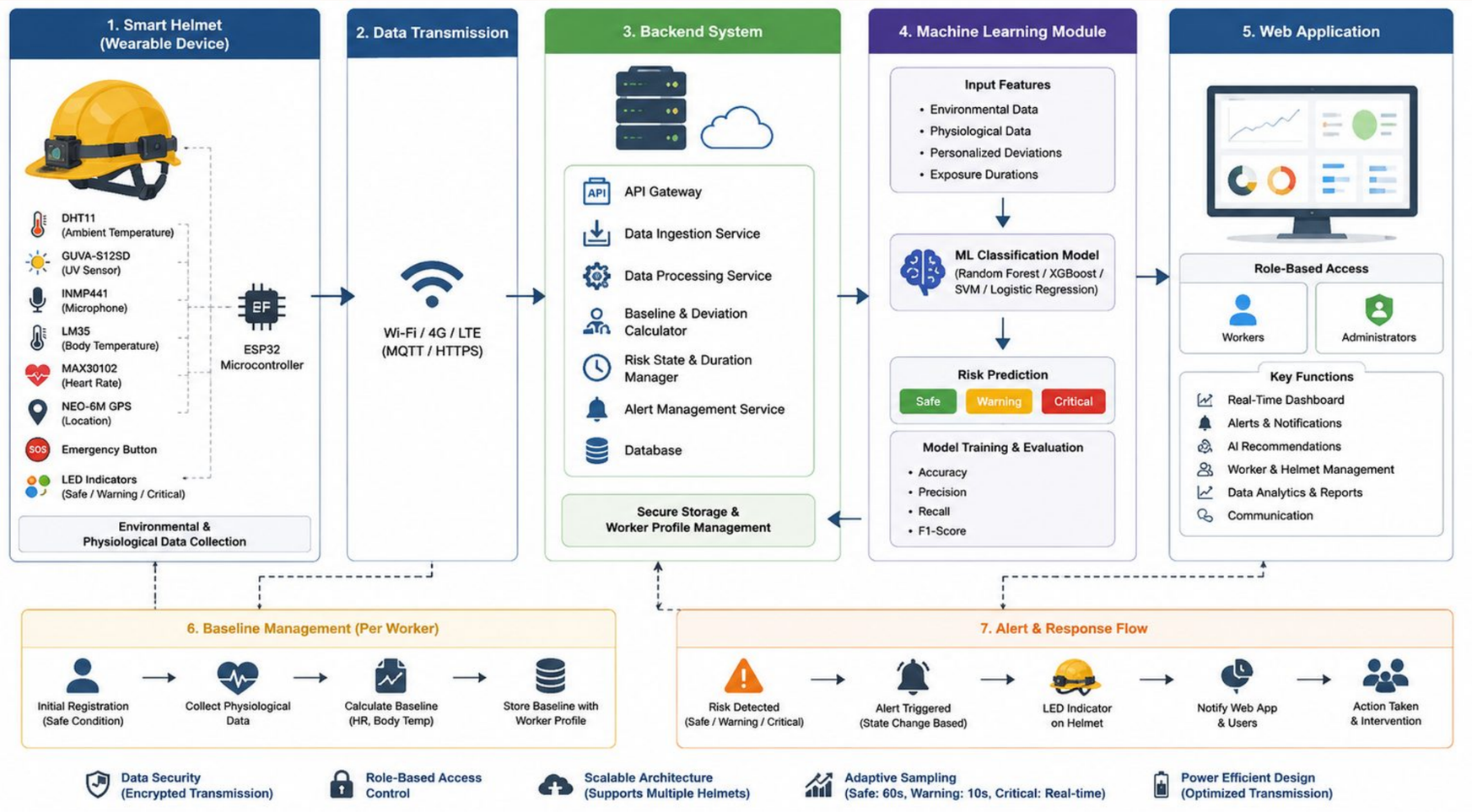
- Receive real-time safety monitoring and emergency alerts.
- Get early warnings and safety recommendations during hazardous conditions.



Safety Officers / Supervisors

- Monitor worker conditions in real time through a centralized dashboard.
- Improve incident response and reduce reliance on manual supervision.

Solution Architecture - “ProSafe” Smart Helmet



Research Aim

Develop an intelligent wearable safety monitoring system for construction workers that continuously analyzes environmental and physiological conditions using machine learning to detect personalized risk levels and generate real-time alerts and safety recommendations.



Multi-Modal Safety Intelligence Component

- IT Number: IT21200892
- Name: Kariyapperuma M.V.

Work Breakdown

01 Research Planning & Requirement Analysis ✅

✅ Wearable Smart Helmet Development 02

✅ Data Collection & Preprocessing 03

✅ Machine Learning Model Development 04

½ Backend System Development 05

06 Web Application Development

07 Testing & Documentation

“ProSafe” Device Implementation

Sensors Used

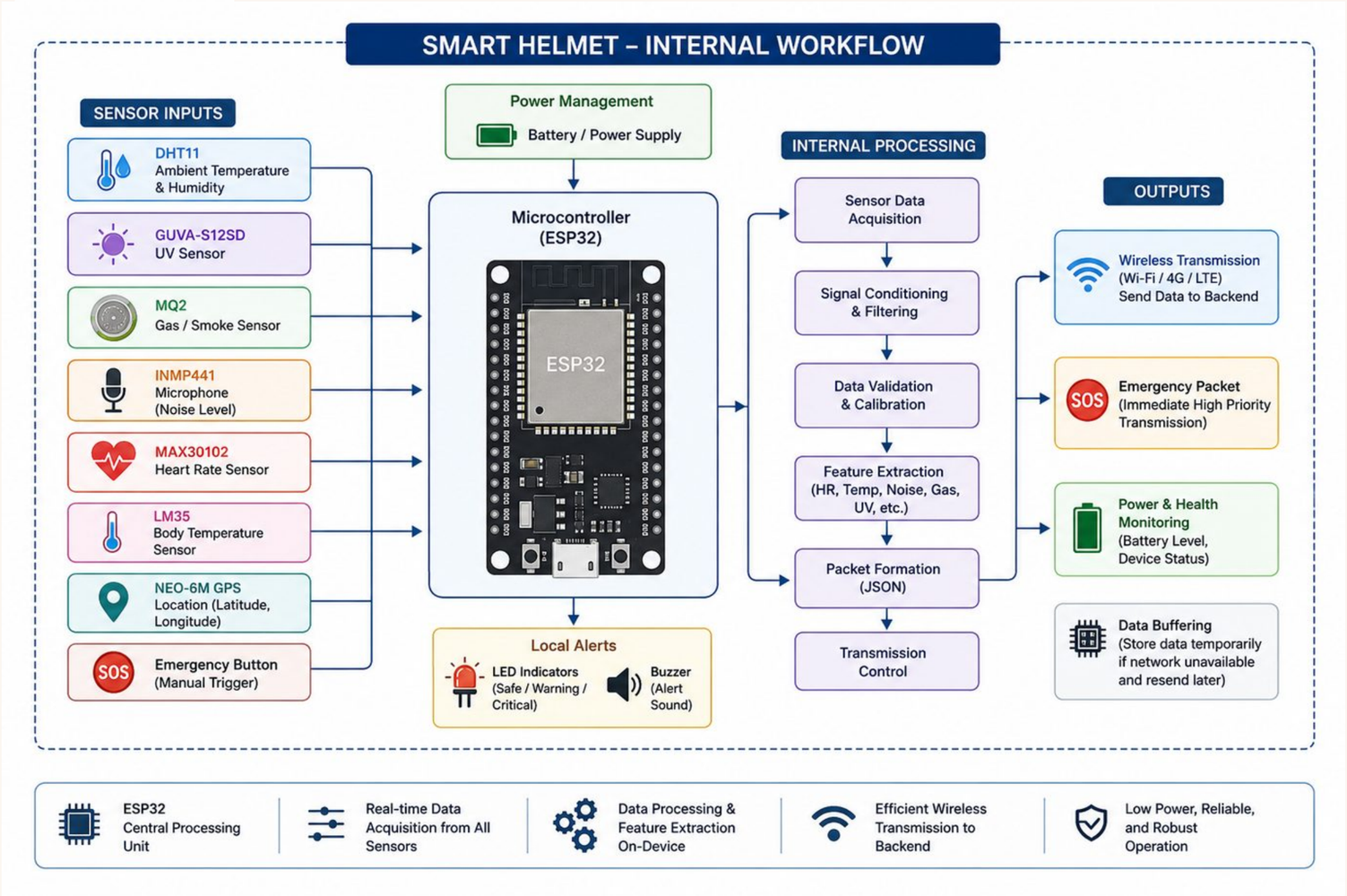
Ambient Temperature	:	DHT11
UV exposure	:	GUVA-S12SD UV detection sensor module
Gas/ VOC	:	MQ2 Gas sensor
Noise	:	INMP441 MEMS microphone
Body Temperature	:	LM35
Pulse Rate	:	MAX30102
Location	:	Neo-6M GPS module
Processing and Control Unit	:	Node MCU ESP32 microcontroller
Battery	:	Lithium-ion battery unit

Steps followed:

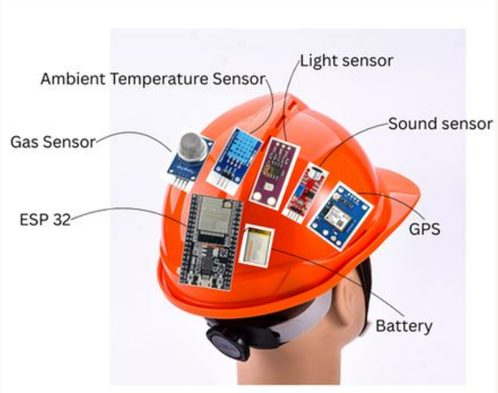
1. Hardware Integration
2. Firmware & Communication Development
3. Sensor Calibration & Testing
4. Real-Time Monitoring Implementation

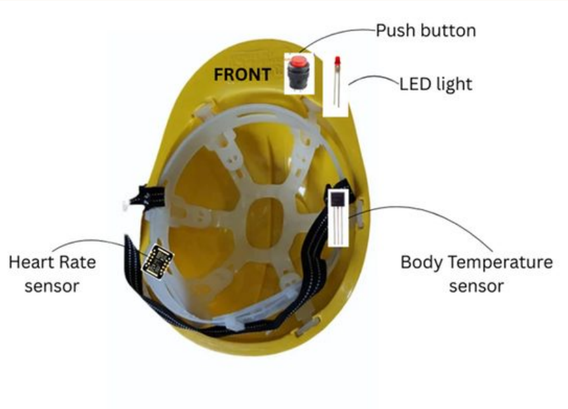
- Normal Packet sending interval: 60s
- Emergency packet sending: Immediately

“ProSafe” Device Implementation



Helmet Design



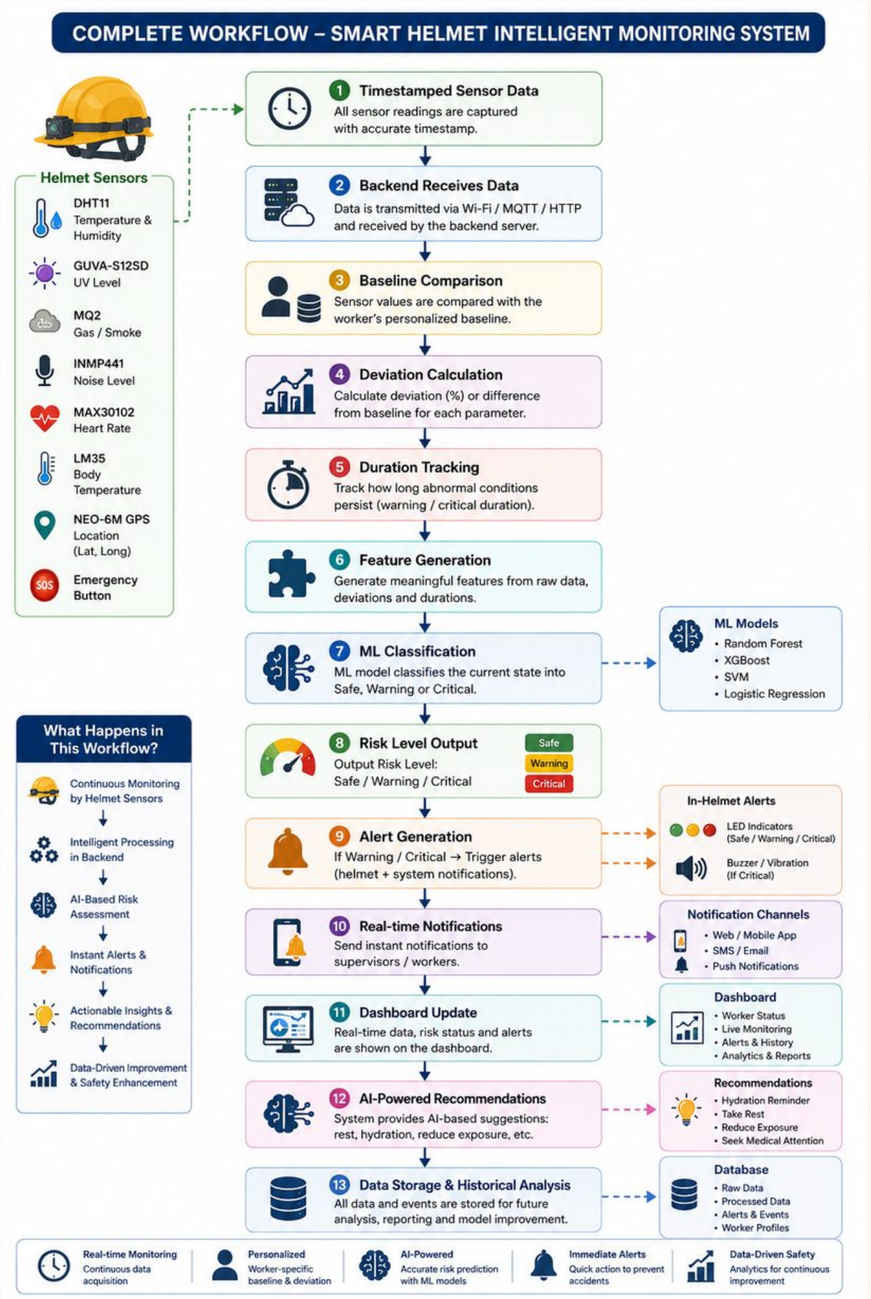


Threshold Values for Environmental Data

- Environmental conditions are evaluated using predefined occupational safety thresholds.

Baseline Comparison

- Each worker has a personalized physiological baseline created during the initial registration phase.
- Current heart rate and body temperature values are compared against the worker's normal baseline values.
- Enables worker-specific safety monitoring instead of fixed physiological thresholds.



Deviation Calculation

- Calculates how much the current physiological readings differ from the personalized baseline.
- Deviation percentage is used to support the classification of Safe, Warning, and Critical states.

[(Current - Baseline)/Baseline]*100

Duration Tracking

- Tracks how long abnormal conditions persist continuously.
- Prevents false alarms caused by temporary fluctuations or physical activity spikes.

Ambient Temperature				
	Safe	**Warning**	**Critical**	**Standard**
Threshold Value	<27 °C	27 - 35°C	> 35 °C	OSHA - NIOSH
Action	• LED – green	• LED – yellow • Waning alert	• LED – red • Critical alert	

Gas Sensor				
	Safe	**Warning**	**Critical**	**Standard**
Threshold Value	<150 ppm	150-300 ppm	>300 ppm	NIOSH fire/smoke early warning guidance
Action	• LED – green	• LED – yellow • Warning alert	• LED – red • Critical alert	

Sound Sensor				
	Safe	**Warning**	**Critical**	**Standard**
Threshold Value	<80 dB	80-85 dB	85-90dB	OSHA PEL
Action	• LED – green	• LED – yellow • Waning alert	• LED – red • Critical alert	

Light Sensor				
	Safe	**Warning**	**Critical**	**Standard**
Threshold Value	<3 UV index	3-8 UV index	>8 UV index	WHO
Action	• LED – green	• LED – yellow • Warning alert	• LED – red • Critical alert	

Machine Learning Model Implementation

Dataset

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	worker_id	timestamp	ambient_temp_c	uv_index	gas_ppm	noise_db	body_temp_c	heart_rate_bpm	baseline_hr_bpm	baseline_body_temp_c	hr_deviation_pct	body_temp_deviation_pct	risk_level
2	W046	1/6/2025 8:00	18.84	1.41	142.39	54.53	37.14	63.2	68	37.14	-7.06	0	Safe
3	W049	1/6/2025 8:00	19.47	0.97	81.33	57.98	35.79	76.3	78.1	36.28	-2.3	-1.35	Safe
4	W060	1/6/2025 8:01	39.71	2.46	112.87	72.53	37.08	85	87.3	36.62	-2.63	1.26	Critical
5	W090	1/6/2025 8:01	19.34	1.22	34.77	53.97	34.79	76.6	88.4	35.41	-13.35	-1.75	Safe
6	W060	1/6/2025 8:02	20.94	2.03	130.74	66.09	35.93	79.4	87.3	36.62	-9.05	-1.88	Safe
7	W048	1/6/2025 8:02	27.39	2.31	84.17	66.81	37.25	99.6	84.3	36.48	18.15	2.11	Warning
8	W071	1/6/2025 8:03	23.3	2.53	104.68	55.94	35.02	122.7	89.1	35.76	37.71	-2.07	Critical
9	W071	1/6/2025 8:03	22.81	2.34	65.9	73.25	35.75	121.5	89.1	35.76	36.36	-0.03	Critical
10	W001	1/6/2025 8:04	22.46	2.37	128.64	66.85	32.96	80.3	74.4	37.85	7.93	-12.92	Critical

+20,000 Records

Model Training; 80%
Model Testing: 20%

Risk Threshold

Safe State

- Physiological deviation below 15%,
- Environmental conditions within acceptable ranges.

Warning State

- Physiological deviation remains between 15%-30%,
- Prolonged environmental hazard exposure occurs.

Critical State

- Physiological deviation exceeds 30%,
- Severe environmental and physiological stress occur together.

Machine Learning Model Implementation

Why These Algorithms Were Chosen,

Random Forest

- Handles noisy sensor data effectively
- Learns complex relationships between environmental and physiological features
- Suitable for structured sensor datasets

XGBoost

- Provides high classification accuracy
- Efficient for tabular data and feature-based learning
- Handles combined risk patterns effectively

Support Vector Machine (SVM)

- Effective for multiclass classification problems
- Performs well with moderate-sized datasets
- Suitable for clear risk boundary identification

Logistic Regression

- Simple and computationally efficient baseline model
- Easy to interpret and compare with advanced models
- Useful for evaluating overall model improvements

Best Algorithm: XGBoost

```
============================================================
TRAINING & EVALUATING MODELS
============================================================

Training Random Forest...
  Accuracy  : 0.9755
  Precision : 0.9773
  Recall    : 0.9755
  F1-Score  : 0.9755

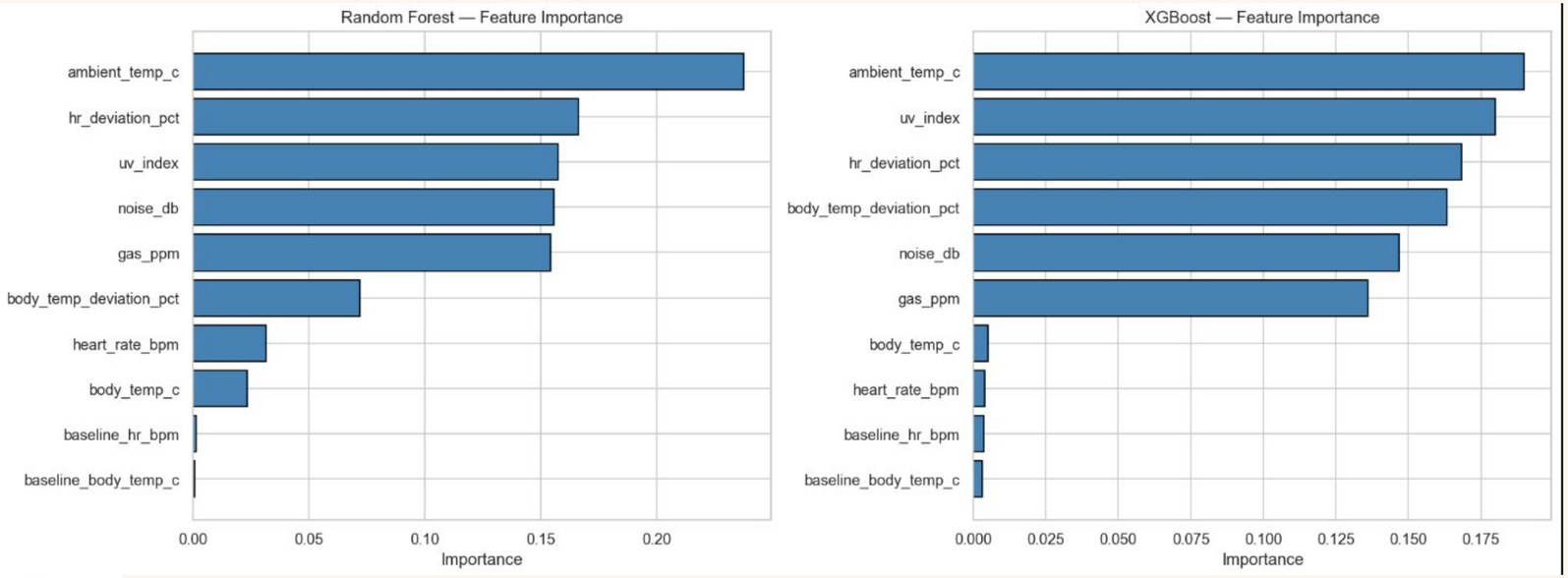
Training XGBoost...
  Accuracy  : 0.9890
  Precision : 0.9890
  Recall    : 0.9890
  F1-Score  : 0.9890

Training SVM...
  Accuracy  : 0.9255
  Precision : 0.9264
  Recall    : 0.9255
  F1-Score  : 0.9258

Training Logistic Regression...
  Accuracy  : 0.6408
  Precision : 0.6435
  Recall    : 0.6408
  F1-Score  : 0.6417
```

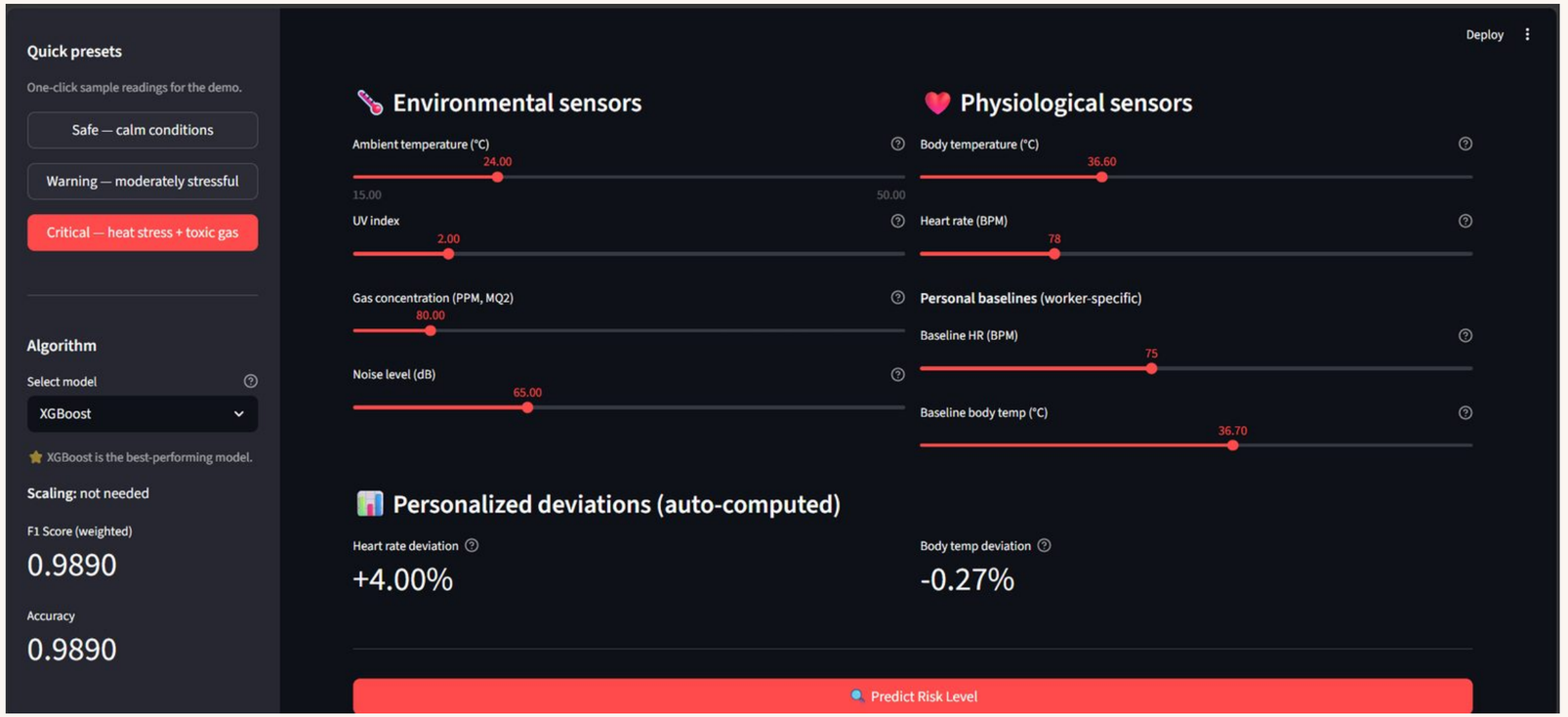
Machine Learning Model Implementation

```
5-FOLD STRATIFIED CROSS-VALIDATION
============================================================
  Random Forest          F1 = 0.9786 +/- 0.0043  (min=0.9753, max=0.9870)
  XGBoost                F1 = 0.9873 +/- 0.0014  (min=0.9857, max=0.9895)
  SVM                    F1 = 0.9262 +/- 0.0027  (min=0.9238, max=0.9315)
  Logistic Regression    F1 = 0.6372 +/- 0.0101  (min=0.6245, max=0.6513)
```



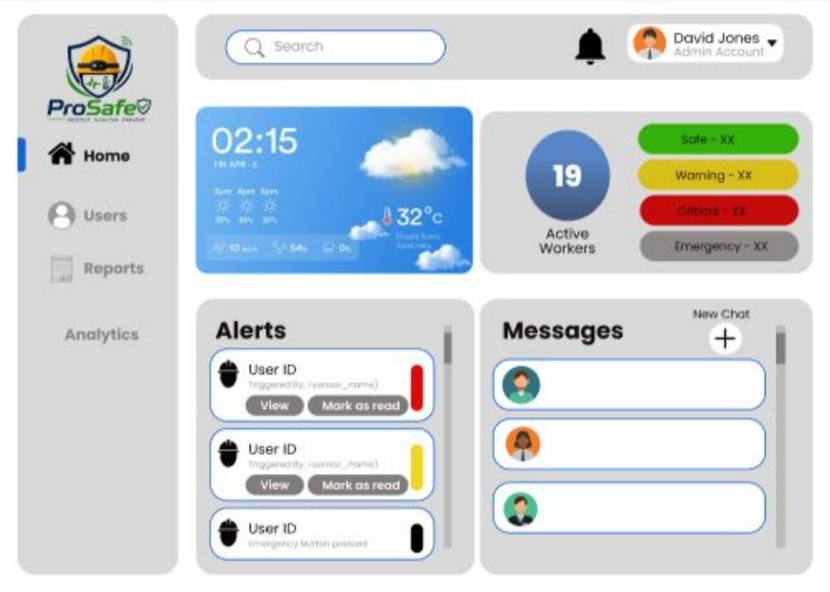
Feature Importance Distribution

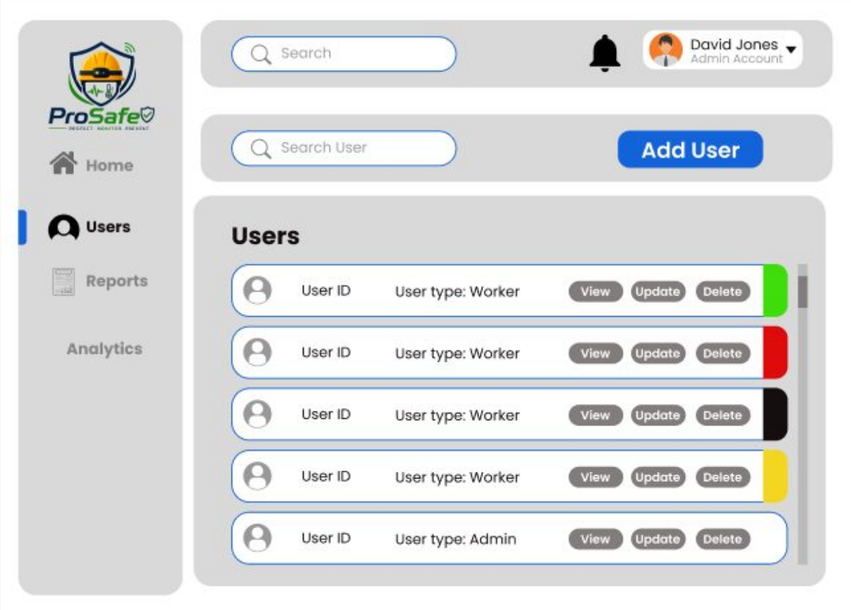
Machine Learning Model Implementation



Streamlit UI - For Demo

“ProSafe” Web Application: Prototype







Figma Prototype Link: https://www.figma.com/proto/82SFOn9eO4nifKZzjmL8Wb/Untitled?node-id=0-1&t=g9Uiz5yDWhazf93k-1

Figma Dev Mode Link: https://www.figma.com/design/82SFOn9eO4nifKZzjmL8Wb/Untitled?node-id=0-1&m=dev&t=g9Uiz5yDWhazf93k-1

Remaining work for PP2

- Backend development
- Web Application
- Testing
- Documentation



Thank you

